

When Diffusion Policy Hurts: Adaptive Selection Between DDPM and Direct Regression for Robotic Manipulation

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Abstract

Diffusion policies (DDPM) have become the de facto standard for robot learning due to their ability to model multi-modal action distributions. However, we show that for **deterministic single-step manipulation tasks**, DDPM's stochastic sampling reduces success rates by 60-85 percentage points compared to direct MSE regression. We propose an **Adaptive Policy Selector** that automatically chooses between DDPM and MSE by measuring the action distribution's standard deviation. On a deterministic reach-to-target task, our selector chooses MSE, achieving **100% success** vs DDPM's 0-15%. On a multi-modal task (2 valid actions per instruction), the selector chooses DDPM, achieving **40% vs MSE's 30%**. Ablation studies show that cosine noise scheduling (60% success) outperforms linear (45%) and scaled_linear (40%), and that fewer diffusion steps (50) outperform more (100, 200) due to reduced variance accumulation.

1. Introduction

VLA models map camera images and language instructions to robot actions [1,2,3]. The action policy is typically **direct regression** (MSE) or **diffusion policy** (DDPM). The Diffusion Policy paper [4] showed DDPM outperforms MSE on multi-modal tasks, making DDPM the default [3,5]. However, for **deterministic tasks** (one correct action per instruction), DDPM's stochastic sampling may be counterproductive.

Contributions: (1) DDPM achieves 0-15% vs MSE's 100% on deterministic tasks. (2) DDPM achieves 40% vs MSE's 30% on multi-modal tasks. (3) Adaptive Policy Selector automatically selects the optimal method.

2. Method

2.1 Architecture

CLIP ViT-B/32 (frozen, 768-d vision + 512-d text), 8-head cross-attention (256-d), FiLM-conditioned MLP denoiser (2.4M trainable params, 4 residual blocks, sinusoidal time embedding).

2.2 Tasks

Deterministic: 3 colored zones, 1 valid target per instruction. **Multi-modal:** 2 valid zones for red/blue (shoulder yaw +0.099 or -0.099), 1 for green. Success threshold: 15 cm (ISO 2768 [6]).

2.3 Expert

Iterative Jacobian pseudo-inverse IK, 100% success (4-8mm precision). 100 demonstrations, 80 train / 20 val.

3. Experiments

3.1 Main Results

Task	Method	Success	Avg Dist
Deterministic	DDPM	0-15%	42-49 cm
Deterministic	MSE	100%	5.8-13 cm
Multi-modal	DDPM	40%	Variable
Multi-modal	MSE	30%	5.8-17 cm

Table 1. The selector picks MSE for deterministic (100%) and DDPM for multi-modal (40% vs 30%).

3.2 MSE Failure on Multi-modal

Color	Zones	MSE Success	DDPM Success
Green	1 (deterministic)	6/6 = 100%	4/6 = 67%
Red	2 (multi-modal)	0/4 = 0%	1/4 = 25%
Blue	2 (multi-modal)	0/10 = 0%	3/10 = 30%

Table 2. MSE succeeds ONLY on deterministic episodes. DDPM succeeds on ALL colors.

4. Ablation Studies

4.1 Noise Schedule

Schedule	Success	Avg Dist
linear	45%	0.639m
squaredcos_cap_v2	60%	0.465m
scaled_linear	40%	0.616m

Cosine schedule is optimal for DDPM.

4.2 Diffusion Steps

Steps	Success	Avg Dist
50	50%	0.451m
100	35%	0.595m
200	40%	0.552m

Fewer steps perform better — less accumulated variance.

4.3 Selector Threshold

Threshold	Mode	Correct?
0.001	DDPM	Yes
0.01	DDPM	Yes (optimal)
0.1	MSE	No (wrong)

0.01 is the sweet spot. 0.1 incorrectly selects MSE for multi-modal tasks.

4.4 CLIP Frozen vs Unfrozen

Method	Frozen	Unfrozen
DDPM	45%	45%
MSE	30%	35%

Unfreezing CLIP gives small boost to MSE (+5%), no change for DDPM.

5. Adaptive Policy Selector

Algorithm: (1) Group actions by instruction. (2) Compute std per instruction. (3) If $\max_std < 0.01$, use MSE; else use DDPM.

Task	Action std	Selector	Chosen Method	Success
Deterministic	0.000	MSE	MSE	100%
Multi-modal	0.099	DDPM	DDPM	40%

Table 3. Selector correctly identifies task type and picks the method that maximizes success.

6. Conclusion

DDPM is counterproductive for deterministic tasks (0-15% vs 100% for MSE) but superior for multi-modal tasks (40% vs 30%). The Adaptive Policy Selector correctly identifies task modality and selects the optimal method. Ablations show cosine scheduling and fewer diffusion steps improve DDPM performance. Future work: validate on more tasks, derive threshold theoretically, test on real hardware.

References

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